Program

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>
#include <linux/if_packet.h>
#include <linux/if_ether.h>
#include <netinet/ip.h>
#define BUFFER_SIZE 65536
int main()
{
    int sock_raw;
    unsigned char *buffer;
    int packet_count = 0;
    sock_raw = socket(AF_PACKET, SOCK_RAW, htons(ETH_P_ALL));
    if (sock_raw < 0)
    {
        perror("Socket Error");
        return 1;
    }
    buffer = (unsigned char *)malloc(BUFFER_SIZE);
    printf("Packet Capture Started...\n");
    while (1)
    {
        int data_size;
        data_size = recvfrom(sock_raw, buffer, BUFFER_SIZE,
                             0, NULL, NULL);
        if (data_size < 0)
        {
            perror("Receive Error");
            break;
        }
        struct iphdr *ip_header =
            (struct iphdr *)(buffer + 14);
        struct sockaddr_in src, dst;
        src.sin_addr.s_addr = ip_header->saddr;
        dst.sin_addr.s_addr = ip_header->daddr;
```

```
        printf("\nPacket No: %d\n", ++packet_count);

        printf("Packet Size : %d bytes\n", data_size);

        printf("Source IP : %s\n", inet_ntoa(src.sin_addr));

        printf("Destination IP : %s\n", inet_ntoa(dst.sin_addr));

        printf("--------------------------\n");

    }

    close(sock_raw);

    free(buffer);

    return 0;

}
```

Output

```
Packet Capture Started...

Packet No: 1
Packet Size : 138 bytes
Source IP : 192.168.1.1
Destination IP : 192.168.1.4
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Packet No: 2
Packet Size : 42 bytes
Source IP : 195.144.192.168
Destination IP : 1.1.152.254
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Packet No: 3
Packet Size : 110 bytes
Source IP : 1.16.143.106
Destination IP : 39.203.164.237
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Packet No: 4
Packet Size : 110 bytes
Source IP : 64.0.0.1
Destination IP : 0.0.0.0
```